Automated Microdrama Promo Discovery & Ranking System

Solution Document (Problem 1) • Show: *I Married a Broken Soldier. He was a Duke* (50 Episodes) • Harman Singh

1. Problem Definition & Editorial Assumptions

Business Objective: Microdrama platforms acquire the majority of new viewers via 20–60s vertical clips on TikTok, YouTube Shorts, and Instagram Reels. Manually analyzing 50+ screenplay episodes per series is an expensive editorial bottleneck. This pipeline automates the extraction and ranking of high-converting promo moments from timecoded scripts with cut-boundary safety and cold-audience viability.

- **Timestamp Fidelity:** Production script timecodes correspond accurately to master video shot transitions.
- **Safe Cut Invariant:** CAMERA shot-start events represent clean visual cut boundaries without tearing audio.
- **Dialogue Primacy:** Spoken dialogue provides primary emotional signals; visual-only actions receive safety weighting.
- **Top-of-Funnel Conversion:** Promos must terminate on unresolved open loops; end-of-series spoilers are penalized.

2. End-to-End Pipeline: AI Perception + Deterministic Enforcement

Production reliability is achieved by strictly separating **creative narrative search (LLM)** from **mechanical rule enforcement (Code)**:

Stage	Component / Owner	Role, Logic & System Invariants
1. Parse	Deterministic Code promo/parser.py	Ingests screenplay blocks (CAMERA, ACTION, SOUND FX, DIALOGUE). Assigns immutable, sequential IDs (e.g. E3-41) and indexes camera shot boundaries.
2. Propose	LLM (gemini-3.5-flash-lite) promo/llm.py	Examines 2-episode sliding windows. Identifies dramatic conflicts and multi-segment candidate clips. **Core Constraint: Model outputs ONLY event IDs; timestamps are forbidden**.
3. Validate & Repair	Deterministic Code promo/validate.py	Resolves IDs to precise times; snaps to shot cuts; **merges contiguous touching segments**; expands cuts that split spoken lines; **repairs near-miss clips** (extends 15–20s to shot cut, trims 60–90s down to ~55s); rejects unrepairable outliers.
4. Cold View	Secondary LLM Pass promo/llm.py	Evaluates standalone clarity (1–5) and confirms unresolved tension without lore. (Disabled in final scoring to eliminate ranking bias between checked and unchecked clips).
5. Rank	Deterministic Code promo/rank.py	Calculates weighted composite score; eliminates near-duplicates (Temporal IoU > 0.40); enforces diversity cap of max 2 promos per episode in top cohort.

3. Multi-Factor Scoring & Ranking Formulation

Composite Score = 0.40 × Hook + 0.40 × Cliffhanger + 0.15 × Standalone + 0.10 × EarlyEpisode − 0.15 × SpoilerPenalty + 0.15 × TailBonus

- **Hook (40%):** First 3 seconds dictate swipe rate. Scores physical actions (slaps, grabs) and high-voltage betrayal lines.
- **Cliffhanger (40%):** Terminal second cuts on unfulfilled questions or threats, never after an apology or resolution.
- **Standalone Clarity (15%):** Validates that a stranger understands character stakes without background lore.
- **Calibrated Tail Bonus (+0.15):** Recalibrated from 0.40 to 0.15 so viral hook and cliffhanger quality drive ranking rather than scene position.

Candidate: Harman Singh (bhanguharmansingh88@gmail.com)

4. Quantitative Telemetry & Ground-Truth Benchmark (from Raw Outputs)

The full pipeline was executed across all 50 screenplay episodes of *I Married a Broken Soldier. He was a Duke*. All metrics below reflect exact values from run_meta.json and output files:

Total Episodes Analyzed 50 Episodes (Full Series)	**Total LLM API Calls** 49 Sliding-Window Calls	**Candidate Proposals** 148 Raw Proposals	**Valid Clips Accepted** 137 Clips (92.6% yield)
Rejected / Discarded 11 Outliers (7.4%)	**Pipeline Execution Time** 1111.6s (~18.5 min total)	**Average Top 10 Duration** 40.8s (range: 25.3s–57.2s)	**Ground-Truth Recall** **8 / 15 (53.3%)** via eval.py

Top Discovered Promo Clips (from promos_ranked.json):

Rank / Target	Dur.	First 3s Hook Dynamic	Terminal Cliffhanger Cut
#1 Ep3 (P066, 01:06–02:03)	57.2s	Scarred stranger strides down wedding aisle; orders guards to unhand bride.	Stranger drops to one knee offering ring; cuts on bride's shocked reaction.
#2 Ep15 (P013, 00:46–01:26)	39.5s	Thug violently shoves bandaged woman to wet ground outside ruined garden.	Alexander steps in front to fight; cuts on high tension before brawl erupts.
#3 Ep5 (P102, 00:54–01:41)	47.5s	Stepmother snatches dead mother's heirloom ledger and flings it into mud.	Dangerous new husband glares down ex-fiance; cuts before retaliation.
#4 Ep1 (P000, 00:09–00:43)	34.1s	Matriarch cold line: *"Gerald is no longer waiting for you."*	Sister announces she is marrying groom instead; cuts on Juliana's gasp.

5. Production Economics & Editorial Workflow

- **Token & Cost Efficiency (Estimated):** A 50-episode season consumes ~350k input tokens. At Flash-tier pricing, total inference cost is estimated at under $0.05 per 50-episode show, fitting entirely within free-tier quotas.
- **Scalability:** Independent show pipelines allow parallel cloud execution; a batch of titles can process concurrently across asynchronous workers.
- **Human-in-the-Loop Efficiency:** Eliminates manual scanning. Editors review the top 10 ranked cards in the Streamlit UI, reducing human turnaround from hours of reading to a brief review per title.

6. Tech Stack & Critical AI Failure Analysis (Grading Requirement)

Implementation Stack: Python 3.11, OpenAI SDK, Streamlit Community Cloud, PyYAML, ReportLab. *Note: This run utilized gemini-3.5-flash-lite; swapping in a stronger flagship model (e.g., Gemini 2.5 Pro) is supported via config.yaml without code modifications.*

Real-World Failure Observed in Run: LLMs exhibit severe *duration myopia* when estimating duration from screenplay timecodes. For example, in candidate **P006** (Ep4), the AI identified the narrative climax but proposed an entire **151.4-second continuous sequence** (>2.5× the 60s ceiling), which would cause instant rejection on TikTok. In candidate **P013**, it selected a single line of dialogue lasting only **19.5s** (<20s ceiling). Furthermore, the model proposed contiguous touching cuts (e.g., 00:02–00:15 and 00:15–00:25) as separate artificial segments.
Defensive Engineering Remediation: Rather than discarding promising dramatic scenes, promo/validate.py was enhanced to: (1) automatically merge contiguous touching cuts, (2) extend near-miss short clips (15–20s) to the next camera shot cut, and (3) trim overlong clips (60–90s) to clean shot boundaries (~55s). This rescued the core Ep3 stranger proposal into a compliant **57.2s clip (#1 ranked)** and boosted valid candidate yield from 57% to 92.6% without human intervention.